Robot framework introduction

Source: https://www.slideshare.net/slideshow/robot-framework-introduction/1604289

An introduction to

Robot Framework

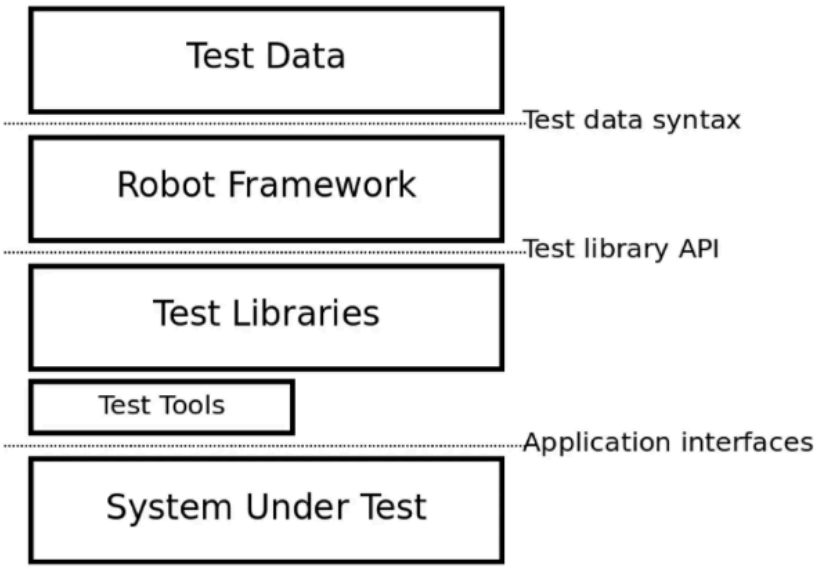
http://robotframework.org

© Copyright Nokia Siemens Networks
Creative Commons Attribution 3.0 License

Fast facts

- Generic test automation framework
  - Utilizes the keyword-driven testing approach
  - Suitable for both “normal” test automation and ATDD
- Implemented with Python
  - Runs also on Jython (JVM) and IronPython (.NET)
  - Can be extended natively using Python or Java
  - Other languages supported via a remote interface
- Open source
  - Hosted on GitHub, Apache 2 license
  - Sponsored by Nokia Networks
  - Rich ecosystem and very active community

High level architecture



Simple keyword-driven syntax

```
*** Test Cases ***
Valid Login
    Open Browser To Login Page
    Input Username    demo
    Input Password    mode
    Submit Credentials
    Welcome Page Should Be Open
    [Teardown]    Close Browser
```

Data-driven tests

```
*** Test Cases ***    USER NAME        PASSWORD
Invalid Username      invalid          ${VALID PWD}
Invalid Password      ${VALID USER}    invalid
Both Invalid          invalid          whatever
Empty Username        ${EMPTY}         ${VALID PWD}
Empty Password        ${VALID USER}    ${EMPTY}
Both Empty            ${EMPTY}         ${EMPTY}
```

Gherkin syntax

```
*** Test Cases ***
Valid Login
    Given browser is opened to login page
    When user "demo" logs in with password "mode"
    Then welcome page should be open
```

Higher level keywords

```
*** Keywords ***
Open Browser To Login Page
    Open Browser    ${LOGIN URL}    ${BROWSER}
    Maximize Browser Window
    Set Selenium Speed    ${DELAY}
    Login Page Should Be Open

Login Page Should Be Open
    Title Should Be    Login Page

Input Username
    [Arguments]    ${username}
    Input Text    username_field    ${username}

Input Password
    [Arguments]    ${password}
    Input Text    password_field    ${password}
```

Simple test library API

```
class Selenium2Library:

    def input_text(self, locator, text):
        """Types given `text` into text field `locator`."""
        print "Typing text '%s' into '%s'." % (text, locator)
        element = self._element_find(locator)
        element.clear()
        element.send_keys(text)

    def title_should_be(self, title):
        """Verifies that current page title equals `title`."""
        actual = self.get_title()
        if actual != title:
            raise AssertionError("Title should have been '%s' but was '%s'."
                                 % (title, actual))
        print "Page title is '%s'." % title


# Above is real Selenium2Library code but slightly simplified.
```

Variables

- Easy to create:

```
*** Variables ***
${BROWSER}        Firefox
${HOST}           localhost:7272
${LOGIN URL}      http://${HOST}/
${WELCOME URL}    http://${HOST}/welcome.html
```

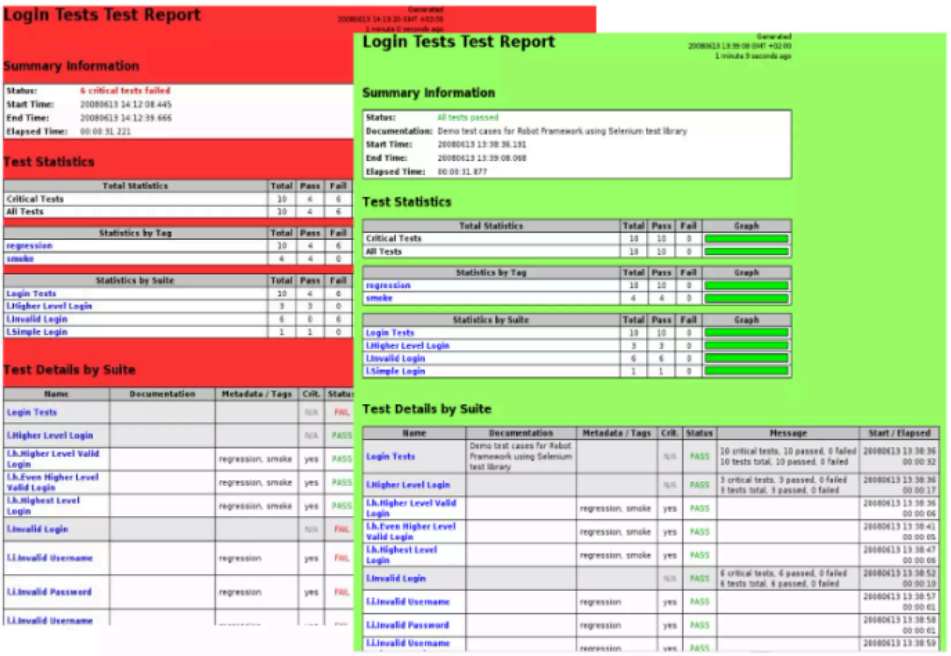
- Override from the command line:

```
--variable BROWSER:IE
```

Tagging

- Free metadata to categorize test cases
- Statistics by tags collected automatically
- Select test cases to be executed
- Specify which test cases are considered critical

Clear reports



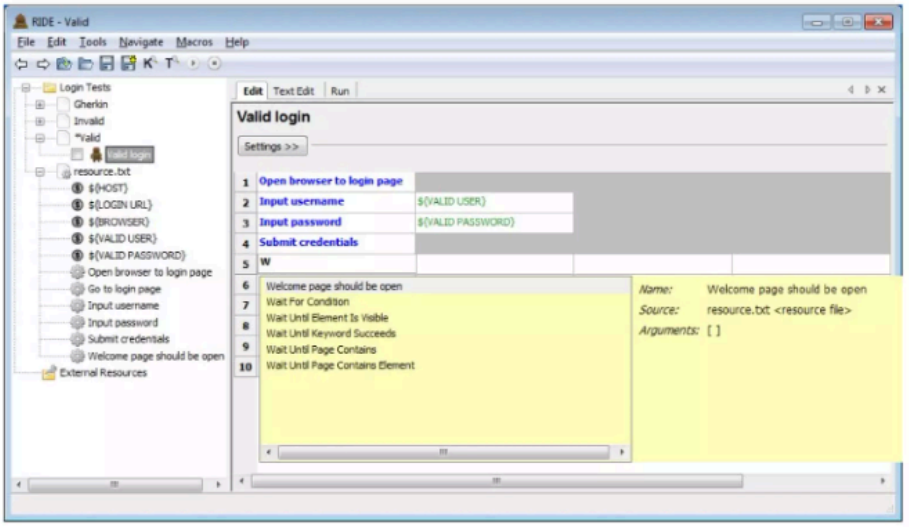
Detailed logs

Different test libraries

- Standard libraries
  - Included in normal installation
  - OperatingSystem, Screenshot, String, Telnet, XML, …
- External libraries
  - Must be installed separately
  - Selenium2Library, SwingLibrary, DatabaseLibrary, AutoItLibrary, SSHLibrary, HTTPLibrary, …
- Project and team specific libraries

Editor support

- RIDE



- Plugins for Eclipse, IntelliJ/PyCharm, SubLime, TextMate, Vim, Emacs, Brackets, Atom, ...

Easy integration

- Test suites are created from files and directories
  - Trivial to store into any version control system
- Simple command line interface
  - Easy to start test execution by external tools
- Output also in XML format
  - All information in machine readable format
  - Outputs from different test runs can be combined
- Plugins for common CI and build tools
  - Jenkins, Ant, Maven

For more information

- Ecosystem front page
  - http://robotframework.org
- Project pages
  - https://github.com/robotframework/robotframework
- Quick Start Guide
  - http://code.google.com/p/robotframework/wiki/QuickStartGuide
- User Guide
  - http://robotframework.org/robotframework/#user-guide
- Demo projects
  - http://robotframework.org/#documentation